

ALEJANDRO ORTIZ PERDOMO

AI Engineer | Machine Learning Engineer

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🌐 alejortizp.github.io/curriculum_HV

PROFESSIONAL SUMMARY

Machine Learning Engineer with end-to-end experience from experimentation to production. Specialized in predictive models (**churn, propensity, NBO**) with quantifiable impact: **Lift 3.0, +15% conversion, +60% contactability**. Proficient in **Scikit-learn, LightGBM, PyTorch** and **Databricks + MLflow** for MLOps. Extensive **deep learning** experience with Transformers and NLP. Combines strong mathematical foundation (Statistics) with modern software engineering to build reliable and scalable ML pipelines.

WORK EXPERIENCE

AI Developer Consultant

Periferia IT

Jun 2025 – Present

- **CV-Job Matching System:** Built complete pipeline with **Azure Document Intelligence + AI Search** processing 500+ CVs/day with 95% extraction accuracy. Semantic search with embeddings reduced screening time by **70%**.
- **Intelligent Automation:** Implemented vectorization and optimized semantic chunking for candidate-job matching, improving recommendation quality by **40%** vs. traditional keyword search.
- **Impact:** System handling **10k+ searches/month** with <200ms latency. Stack: Python, Azure AI Search, Document Intelligence, FastAPI.

EDUCATION

Master in Software Engineering

[Universidad de los Andes](#)
2025 – In Progress

B.S. in Statistics

[Universidad del Valle](#)
2011 – 2016

Data Science Diploma (DS4A)

Correlation One, 2022

Deep Learning Diploma

Univ. Autónoma Occidente, 2023

LANGUAGES



Spanish

Native



English

Professional Proficiency (B2)

KEY SKILLS

Effective Communication

Problem Solving

Analytical Thinking

Team Collaboration

Adaptability

Continuous Learning

Time Management

Technical Leadership

Machine Learning Engineer

Sep 2023 – Jun 2025

Global Hitss

- **GenAI & RAG:** Designed and deployed conversational agent with RAG that reduced support tickets by **40%**, processing **10k+ queries/month** with 85% resolution without escalation. Stack: LangChain, Azure OpenAI, FAISS.
- **MLOps:** Implemented CI/CD pipeline in **Databricks + MLflow** reducing deployment time from 2 weeks to **2 days**. Production models with automatic drift monitoring and retraining.
- **Reinforcement Learning:** Productized *Next Best Offer* (NBO) system with A/B testing that increased conversion by **15%** and revenue by **\$10k per person**.
- **Churn Model:** Developed predictive model with **Lift of 3.0** and 82% recall in top decile, integrated to CRM for proactive retention. Impacted **50k+ customers**.

Machine Learning Consultant

Aug – Sep 2023

Grupo CINTe

- **ML Observability:** Designed comprehensive monitoring methodology for *data/model drift* detection with automated retraining alerts, adopted as the data science team's standard.
- **Optimization:** Audited and optimized LightGBM and Random Forest algorithms, improving computational efficiency and reducing inference times.

Data Scientist

Mar 2023 – Jul 2023

NAE

- **Contactability Model:** Developed **LightGBM** model that increased effective contact rate by **60%** (from 25% to 40%), optimizing time windows and channels. Impacted campaigns with **100k+ contacts/month**.

Data Scientist II

May 2022 – Jan 2023

Xolit

- **Payment Propensity Model:** Developed predictive model for payment probability and portfolio agreement, maximizing collection through **automated management priorities** based on propensity scores.
- **Recommendation System:** Implemented optimal contact time-slot recommendation engine per account holder, increasing debt collection management effectiveness.
- **Customer Segmentation:** Designed clustering-based marketing strategy for account holders, enabling personalized communication per segment to optimize collection rates.
- **Monitoring:** Built **Power BI** dashboards for tracking collection KPIs and production model performance metrics.

CERTIFICATIONS

AI-102: Microsoft Certified Azure AI Engineer Associate

Microsoft, Feb 2026

NLP Specialization (RAG/Transformers)

Coursera, Jun 2025

PyTorch Ultimate

Coursera, Mar 2025

MLOps and Databricks

Netec, Dec 2024

AWS Cloud Practitioner

Udemy, Feb 2022

TECHNICAL SKILLS

(years exp.)

GENAI & LLMS

LangChain 2+ (Agents, RAG)

Azure OpenAI 2+ (GPT-4, Embeddings)

Prompt Engineering 2+

Vector DBs 2+ (FAISS, Pinecone)

AZURE AI & CLOUD

Azure AI Search 1+

Document Intelligence 1+

Databricks + MLflow 3+

ML/DL FRAMEWORKS

PyTorch 2+ | TensorFlow 3+

Scikit-learn 4+ | XGBoost 4+ | LightGBM 4+

Hugging Face Transformers 2+

MLOPS & ENGINEERING

Python 7+ | SQL 7+ | PySpark 3+

Docker 2+ | CI/CD 2+

FastAPI 3+ | Streamlit 3+

Git 4+ | GitHub Actions 2+

OPEN SOURCE PROJECTS



Email Agent with LangChain

— github.com/alejortizp/email-agent

Intelligent agent that queries emails via IMAP, classifies with LLM, and generates contextual automatic responses. Reduces response time by **80%**.

Stack: LangChain, Azure OpenAI, IMAP, Python, Prompt Engineering



NL-to-SQL/Pandas Agent

— github.com/alejortizp/nl-to-code

Agent that translates natural language to SQL queries and Pandas operations. Includes code validation, secure sandbox execution, and automatic results visualization.

Stack: LangChain, SQL, Pandas, Code Interpreter, ReAct Pattern



RAG Restaurant Ordering System

— github.com/alejortizp/rag-ordering

Conversational ordering system with RAG over restaurant menu. Agent processes product selection in natural language, generates invoices, and calculates totals. Includes inventory management.

Stack: LangChain, FAISS, Azure OpenAI, FastAPI, Streamlit